

Sheet 2

1. List the layers of the Internet model.

Physical layer

Data Link layer

Network layer

Transport layer

Application layer

2. Which layer in the Internet model is the user support layer?

The Application layer is the user support layer because it provides services to the user such as e-mail, file transfer, and remote login.

3. What is a peer-to-peer process?

A peer-to-peer process is when corresponding layers in two communicating devices exchange information using protocols of that layer.

Each layer on the sender communicates logically with the same layer on the receiver.

4. What are headers and trailers, and how do they get added and removed?

Header: Extra control information added at the beginning of data by each layer.

Trailer: Extra control information added at the end of data (mostly by the Data Link layer).

Encapsulation: Each layer adds its header (and trailer if needed) before passing to the lower layer.

Decapsulation: Each layer on the receiving side removes its corresponding header/trailer.

5. What are the responsibilities of the data link layer in the Internet model?

Framing: Divides bit stream into frames.

Physical addressing: Adds source/destination MAC addresses.

Flow control: Prevents receiver overload.

Error control: Detects and retransmits lost or damaged frames.

Access control: Controls access to shared media.

6. What are the responsibilities of the transport layer in the Internet model? address?

Process-to-process delivery.

Service-point (port) addressing.

Segmentation and reassembly.

Connection control (connection-oriented or connectionless).

Flow control and error control (end-to-end).

7. Name some services provided by the application layer in the Internet model.

E-mail (SMTP, X.400)

File Transfer (FTP, FTAM)

Remote access / Terminal emulation (TELNET)

Directory services (X.500)

8. How are OSI and ISO related to each other?

ISO (International Organization for Standardization) created the OSI (Open Systems Interconnection) model as a standard framework for communication between systems.

9. One or more layers of the OSI model:

*	Function	OSI Layer
1	Route determination	Network
2	Flow control	Data Link / Transport
3	Interface to transmission media	Physical
4	Provides access for the end user	Application
5	Reliable process-to-process delivery	Transport
6	Route selection	Network
7	Defines frames	Data Link
8	Provides user services (e-mail, file transfer)	Application
9	Transmission of bit stream	Physical
10	Communicates directly with user's app	Application
11	Error correction and retransmission	Data Link / Transport
12	Mechanical, electrical interface	Physical
13	Carrying frames between adjacent nodes	Data Link
14	Format and code conversion	Presentation
15	Establishes, manages, and terminates sessions	Session
16	Ensures reliable data transmission	Transport
17	Log-in and log-out	Session
18	Independence from data representation differences	Presentation

10. In Figure 1, assume that the communication is between a process running at computer A with port address i and a process running at computer D with port address j. Show the contents of packets and frames at the network, data link, and transport layer for each hop.

LAN1: [(42)(40)] [(A)(D)] [Data = (i, j)]

LAN2: [(80)(82)] [(A)(D)] [Data = (i, j)]

11. Compare the OSI model with the TCP/IP model. Show how the layers in the OSI model correspond to the layers in the 4-layer and 5-layer versions of the TCP/IP protocol suite

OSI Model (7 Layers)	TCP/IP Model (5 Layers)	TCP/IP Model (4 Layers)	Description / Function
7. Application	5. Application	4. Application	Provides user services (e-mail, FTP, web, etc.)
6. Presentation	5. Application	4. Application	Data translation, compression, encryption
5. Session	5. Application	4. Application	Dialog control and synchronization
4. Transport	4. Transport	3. Transport	Process-to-process delivery (TCP/UDP)
3. Network	3. Internet	2. Internet	Logical addressing and routing (IP, ICMP, ARP)
2. Data Link	2. Network Access	1. Network Interface	Physical addressing and framing
1. Physical	1. Network Access	1. Network Interface	Transmission of bits over physical medium

12. There are physical address and logic address associated with the communication. Discuss what address changes and what address does not change in this case.

LAN	MAC	IP	DATA
LAN1:	[(33)(10)]	[(A)(B)]	[Data]
LAN2:	[(95)(66)]	[(A)(B)]	[Data]
LAN3:	[(20)(99)]	[(A)(B)]	[Data]

Physical address (MAC):

Used within a local network (LAN) , Changes at each hop.

Logical address (IP):

Used for end-to-end delivery , Does not change during transmission.

13. follow the figure shown below and complete the headers

LAN3: [(31)(8)] [(R)(X)] [Data]

LAN2: [(20)(21)] [(R)(X)] [Data]

LAN1: [(9)(30)] [(R)(X)] [Data]